M0: General Project Requirements

This document outlines expectations for students' projects and presentations, which will be completed in teams.

Project Requirements

Project development and submission must follow specific guidelines depending on the nature of the project. Deviation from these criteria, depending on the requirement and degree, may result in lowered earned score (may even result in a grade of ***zero***).

Projects must, *at minimum*, demonstrate mastery of Computer Engineering topics (due to accreditation requirements). Projects may focus on a mix of software and hardware, but must demonstrate understanding of **both** software and hardware. In the case that "off the shelf" components are used (e.g. libraries, micro-controllers, etc.), ***the project should re-target/re-scope to compensate*** – i.e., these tools allow projects to go further, not to require less effort.

Project Types

Project requirement will vary in part depending on the type of project. Where there is overlap, teams should target the primary objective of the project when considering requirements.

End-User Tools

1. Intended for interaction directly with the end-user
2. Primary purpose is to accomplish one or more tasks (vs. entertainment)
3. Build specifications will vary depending on platform
4. Has target functionality and aesthetic theme

Entertainment

1. Intended for interaction directly with the end-user
2. Primary purpose is entertainment (vs. accomplishing tasks)
3. Build specifications will vary depending on platform
4. Software has a story, mechanics, and aesthetic theme; hardware has justified use-case

Infrastructure

1. Intended for use by engineers (not the end-user)
2. Primary purpose is to facilitate the development of other engineering artifacts
3. Build specifications are ***universal***, regardless of target platform
4. Has an application programming interface (API) and/or physical interface

Build Specifications

The build specifications for projects vary depending on project type and, in some cases, target platform. These build specifications must be met for all project milestones. We will consider exceptions on a case-by-case basis.

1. Software releases should be prepared as a source distribution package **buildable and installable** by instructors via a single command (e.g., `pip install project`, `make`, `cargo build`) on Windows 10/11, Ubuntu 20.04 (native/WSL), Android 16, iOS, or MacOS Tahoe. Teams must also provide binary distributions for their supported platforms. Full points may only be earned on if both the build process and binary executable function properly.
2. Hardware releases should provide detailed schematics and other design documents **sufficient to completely understand (i.e. *replicate* exactly)** the device, along with a physical artifact.
3. Infrastructure projects must include at least one **substantial and illustrative** sample use case (application/device) to demonstrate functionality as proof of concept.

Team Coordination

The following expectations apply to each team and its members' work and coordination with one another.

Team Formation

1. Teams may only work on approved projects.
2. Teams must consist of 3 to 5 members.
3. Team formation is by *consensus* – i.e., a student must receive the consent of **all** existing team members before joining a team.
4. Students, without exception, must join a team by the set deadline. Students, under no circumstances, will be placed into a team by course staff (i.e., it is **solely** on a student to join a team by the designated deadline). **Failure to do so may result in immediate failure of the course.**

Contributions

1. All students must contribute, in approximately equal proportion, to the technical aspects of the project. **Failure to do so may result in points being equitably redistributed based on effort expended (as measured against third-party logs).**
2. Serving as a "team lead" does not absolve or diminish a team member's expectation of (equitable) technical contributions.
3. Team members, "on [their] honor" (see UF Reg. 4.040(3)(c)2 (2023)), are *obligated* to complete peer review documentation **honestly and with candor**.
4. Teams, in their entirety, must meet **weekly** and should keep a meeting log of attendance in third-party source (e.g. GitHub repo). This meeting must be distinct from the stakeholder meeting.

Design Repository

1. Team must use both a third-party repository system (e.g., GitHub) and issue tracking system (e.g., GitHub Issues, Trello). The issue tracking system must include the capability to assign tasks to members, and it must be utilized.
2. Repository commits should be made for all schematics, code, research, and resources developed, without exception. Git LFS support can be used for very large files such as media assets. Logs must be **committed and pushed** regularly to the repository.

   Logs of time spent without supporting evidence is **insufficient** and **does not constitute evidence of effort expended**. Tasks such as research and physical artifact work must be backed by "proofs of work" (e.g. summary of research and findings, pictures of work completed).
3. Any contribution should be placed in the repository, including planning documents that are developed, in order to facilitate determination of effort.
4. Contributions must be committed to the repository **by the individual contributor**; items should be merged within the repository system directly to create a clear trail of effort and contribution.
5. All elements must be merged to `main` by project milestone deadlines.

Presentations

1. All students must contribute to the presentation directly – i.e., everyone must present.
2. All students must take part in the planning and preparation of the presentations to receive credit.
3. All team members should have approximately equal speaking time during the presentations.
4. Presentations should be well-rehearsed and must be **fault-tolerant** (e.g. demos should have backups – e.g. illustrative video recordings).
5. Presentations have strict set time limits. Students may only present during their alloted period. Any content presented outside of a team's alloted period will not be considered for scoring.